Solution:

(i) To buy any one of the new car from the given choices

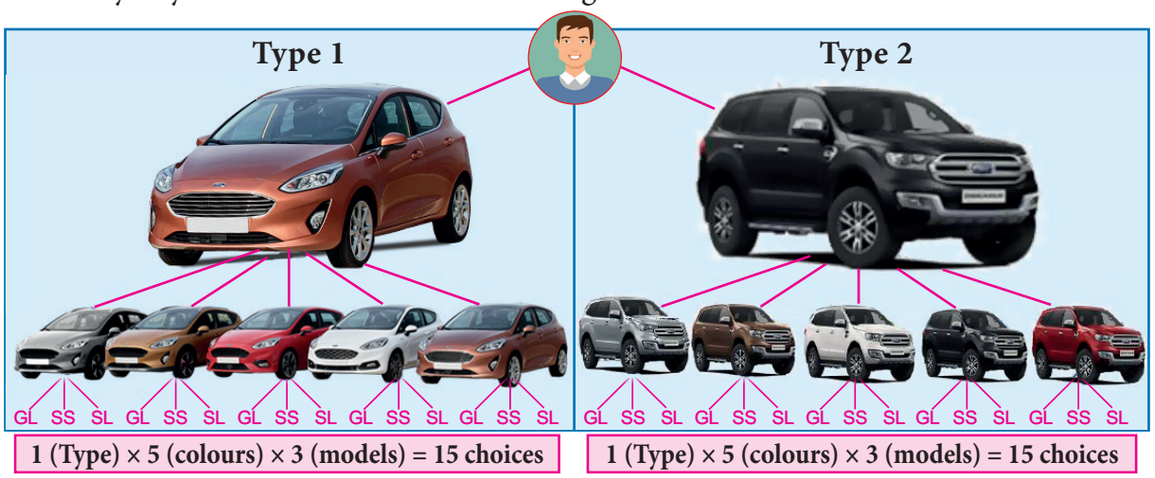


Fig. 7.10

Here, we have **2 types** of car with **5 different colours** and **3 models** in each colour. Therefore, there are **30** **[2 (1 × 5 × 3)]** **different ways** to buy a new car by Madhan.

(ii) If the **white colour** is not available in **Type 2,** then...

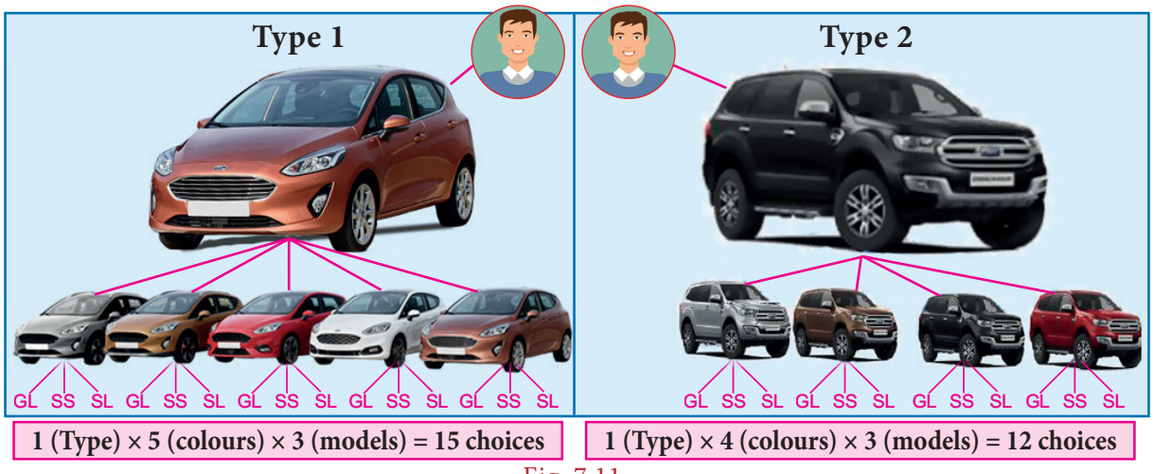


Fig. 7.11

(i) For **Type 1,** we have 5 colours and 3 models and hence there are **1 × 5 × 3 = 15 choices.**

(ii) For **Type 2,** we have only 4 colours and 3 models and hence there are **1 × 4 × 3 = 12 choices.**

Therefore, there are **27** **(15 + 12)** **different ways** to buy a new car by Madhan

The above example illustrates both the addition and multiplication principles.

7.3 SET - Game

Any game that uses features can be used to stimulate logical thinking and it provides an interesting and challenging context for exploring ideas in discrete mathematics.

Now, let us learn about a SET game

A **SET** game proves to be an excellent extension for activities that involve organizing the objects by attributes. The **SET** game builds the cognitive, logical, spatial reasoning as well as visual perception skills.

The **SET** game is a puzzle that uses cards which have four features on them. They are shapes, colours, shades and the number of shapes.

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